

Activity 2 Business Case and Cost-Benefit Analysis with GenAI

Activity 2 | Topic 01 | C013 | Version v1.0

Course: Generative AI for Project Management (C013) Topic 01: Project Initiation and Planning with GenAI Learning outcome: LO1 - Establish the economic feasibility of a project using GenAI-assisted analysis. Tools: GenAI tool | Business Model Canvas Studio (alfredang.github.io/bcm/) | Cost Benefit Analysis.xlsx | Business Case on a Page.docx

Goal

Before a project is authorised somebody has to prove it is worth doing. You will use GenAI to draft a cost-benefit analysis and a one-page business case, then validate the business model behind it on the Business Model Canvas Studio.

What you'll produce

A completed cost-benefit analysis, a one-page business case, and a validated 9-block Business Model Canvas.

Activity files

- [Activity guide PDF](./activity-guide.pdf)
- [Prompt sheet](./prompts.pdf)
- [Detailed instruction manual](./instruction-manual.pdf)
- [Mock data workbook](./mock-data.xlsx)

Core prompt

List the one-off and recurring costs, and the quantified annual benefits, of [your project] over 3 years

Step-by-step

1. Form a group of 3-5. Download 'Cost Benefit Analysis.xlsx' and 'Business Case on a Page.docx' from the course template pack.
2. Agree on a project scenario for your group - for example introducing a GenAI knowledge assistant for your customer service team.
3. Prompt GenAI: 'List the one-off and recurring costs, and the quantified annual benefits, of [your project] over 3 years. Present as a table with a column for how each figure would be verified.'
4. Challenge the output with the Fact Check List pattern - ask it to separate figures it knows from figures it assumed. Replace every assumed figure with your own estimate.
5. Enter the validated figures into Cost Benefit Analysis.xlsx and read off the payback period and net benefit.
6. Open the Business Model Canvas Studio at <https://alfredang.github.io/bcm/> - one member clicks 'Create New Canvas', enters a name and canvas title, then shares the QR code or invite link so the rest of the group joins.
7. Prompt GenAI: 'Draft a Business Model Canvas for [your project]. Give me 2-3 sticky notes for each of the nine blocks: Key Partners, Key Activities, Key Resources, Value Propositions, Customer Relationships, Channels, Customer Segments, Cost Structure, Revenue Streams.'

8. Add the AI's suggestions as sticky notes in the matching blocks. Set each note's status to Draft. Use the (i) 'Show guidance' button on any block you are unsure about.
9. As a group, review every note. Change the status to Validate for notes you have evidence for and Confirmed for notes you are certain of. Delete anything the AI invented that you cannot support.
10. Complete Business Case on a Page.docx using your validated numbers and canvas, and share your findings with the class.

Check your work

Check: Your canvas has all nine blocks populated, every note carries a Draft/Validate/Confirmed status, and your business case states a payback period you can defend with your own figures.

Remember: GenAI gives you a fast first draft. Check every figure, source and assumption before the output leaves your hands - you remain accountable for it.

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